

Assessment 4: Case project part 2

Due date: Week 11: **11:59 AM (midday), Wednesday 29 April**

Project Overview

Now, in Assessment 4, we'll conduct an in-depth analysis to address the food waste issue at CoastalFresh. This assessment mirrors the types of projects you will likely encounter in your future workplace. You will continue with further data analysis using the **same personal Excel data from your Assessment 3**. For instructions on downloading and data descriptions, please refer to **pages 6-7 of the Assessment 3 Guideline document**.

- **Word Limit:** 2,000 words with a 10% buffer, allowing for a maximum of 2,200 words without penalty. There's no minimum word requirement, so any word count below 2000 is acceptable.
- **Word Count Rule** –Everything in your report counts towards the word limit, including headings, subheadings, and in-text citations. The only content that **does not** count towards the word limit is the **reference list** (bibliography) and any text that appears inside inserted screenshots or images. Please use your Word document's built-in word count feature to check your word count accurately.
- **Structure and Format:** No need to include an introduction or executive summary. **Start your report directly by answering Section 1.** No specific format requirement (such as font size, etc.). Just ensure your essay is very clear to read by using formal language, inserting clear diagrams, and applying clear headings and subheadings. It's also okay to use dot points in sections where they help present information more clearly and make your report easier to read. Ensure you include your full name and student ID in the footer of each page.
- **Referencing Style:** A well-chosen reference can enhance the quality of your work and may positively impact your mark. You are encouraged to cite relevant external sources where appropriate to support your analysis. If you do reference, please use the Harvard referencing style. Please use this useful example link - [Cite Sources with Harvard Referencing | UNSW Australia](#)
- **Submission** – **You only need to submit your report as a Microsoft Word document. There is no need to submit your personal Excel file.** Simply insert all your Excel diagrams into your report as screenshots. Please keep your original files properly, as they will also be used in Assessment 4.
- We recommend that you use [UNSW OneDrive](#) to store your document online as a backup, in case you experience any technical issues with your computer.

Section 1 – Statistical Analysis (40%):

This section is approximately 800 words (as a guideline, not a strict limit).

- 1) (a). Using the concepts from lectures and tutorials from Week 5, calculate the **95% confidence intervals** for the **quantity wasted** in Fresh Bakery and Chilled Dairy separately across the entire period in your personal Excel data file. Show your calculation process and explain what these intervals mean. This provides a baseline understanding of the variability of food waste.

(b). Building on part (a), select 2 variables you believe will impact food waste. These variables can come from any source (e.g., real-world factors), not just your Personal Excel data file. For each variable, clearly explain whether the relationship (correlation) is positive (the variable and food waste move in the same direction), negative (the variable and food waste move in opposite directions), or mixed (the relationship may change depending on the situation). You are not required to perform any calculations. Instead, describe how changes in each variable might increase or decrease the overall level or variability of food waste, and how this could affect the 95% confidence intervals from part (a).

(c). Based on the two variables identified in part (b), choose **one of them** and create both a Null Hypothesis and an Alternative Hypothesis about its relationship with food waste. Clearly explain the rationale behind them and discuss how testing this relationship could help determine whether the variable has a meaningful impact on food waste. You are not required to conduct any data analysis or testing; simply justify your hypotheses.

Hint: It'll be helpful to make a few relevant citations to support your reasoning in (b) and (c). We have taught how to make a good citation in our Week 4 tutorials (slides available on Moodle).
- 2) (a). Using your personal Excel data file (the same one used in Assessment 3), as taught in Week 8 tutorial, calculate the **correlation** between "Days_On_Shelf" and "Food Waste" (Quantity wasted column) for Fresh Bakery and Chilled Dairy separately. Present the correlation values in a clear table and include a scatter plot to visualise the relationship in your report. Explain the direction and strength of each correlation, and discuss how "Days_On_Shelf" influences food waste. *(This step examines whether a measurable linear relationship exists between these two variables)*

(b) Please choose **either** Fresh Bakery **or** Chilled Dairy products for your analysis. Using Excel, run a simple linear regression between "On_Shelf_Duration" and Food Waste ("Quantity wasted" column), treating Food Waste (Quantity wasted column) as the dependent variable (Y). Include clear screenshots of your regression Excel output, and provide a written analysis to interpret the results of your regression model, your key findings, and their relevance to food waste. *(This step uses a linear regression to further examine the relationship)*
- 3) Next, using Excel, run a **Multiple Linear Regression** on food waste (Quantity wasted column) for both Fresh Bakery and Chilled Dairy products. You may choose to run separate regressions for each category or combine them into one regression model that includes both. Select "On_Shelf_Duration" as one of your independent variables, and at least 1 or 2 additional variables from your personal Excel data file. You can either use existing variables or create new ones (such as creating a new column) based on the data you already have.

For each additional variable (**no need to explain the On_Shelf_Duration**) included in your

regression model, briefly explain why you are including them (you may build on your earlier discussion in Section 1 to avoid repetition). If you choose to run separate regressions, ensure you use the same 2 or 3 variables for both Fresh Bakery and Chilled Dairy.

Include clear screenshots of your multiple linear regression Excel output, and provide a written analysis interpreting the results of your regression model. Clearly explain the statistical significance of each independent variable, identify the most influential variable, and discuss how including this variable might affect the regression outcomes. (This final step extends your analysis by evaluating multiple factors simultaneously and identifying the key drivers of food waste.)

Hint: Please refer to the Week 8 and 9 tutorials slides (available on Moodle) for more details on **correlation**, **linear regression**, and **guideline/tips for Assessment 4**. You can also find an "Assessment 4 Resource file" on Moodle (located just below this Assessment 4 document), which provides guidance that you can easily use to help you prepare for this assessment.

Section 2 – Ethical Dilemma (25%)

This section is approximately 550 words (guide only, not a word limit).

In Assessment 3, you examined an ethical dilemma regarding CoastalFresh's decision to use discounts to ensure near-expiry products are consumed rather than discarded and discussed how this decision might impact one particular stakeholder.

In this assessment, you will build on that discussion by applying the **7-Step Ethical Decision-Making Framework** taught in Week 4 and revisited in Week 10.

As CoastalFresh is your client, you need to use this "7-step framework" to provide a structured and organised answer that helps CoastalFresh understand the ethical impacts of its discount decision on both the business and its related stakeholders. Your explanation should be practical and easy to understand, so that CoastalFresh can make informed final decisions.

Section 3 – Problem Solving (35%)

This section is approximately 750 words (guide only, not a word limit).

1) At the end of our business report, we need to provide an overview that highlights the key takeaway from the entire report. **Please insert the table provided below directly into your report to answer this question.** Type all your answers as text inside the table (**do not insert this table as a screenshot**). All content within the table will count towards the word limit, so ensure your answers are clear and easy to read.

Developing Practical Solutions	
Situation	
Observation	
Resolution	

Instruction:

For Situation – Provide a clear and concise summary of the key aspects of this case. Your explanation should be simple and straightforward, allowing readers to quickly understand the main points of your entire report (both Assessments 3 and 4). Focus on the critical elements of the case, highlighting the most important details relevant to your analysis and recommendations.

For Observations – Summarize the key insights from your analyses in Assessments 3 and 4. This should provide a quick overview of your main findings, including:

- **Logic Tree Insights:** summarise the key points (no need to discuss the entire tree). You may include a screenshot of your tree, or part of your tree, if it helps support your explanation.
- **Excel Data Findings:** Highlight the significant outcomes from your data analysis. You may attach a screenshot to display key results you wish, such as any Excel diagram you want to show again here (note that content inside screenshots does not count toward the word count).
- **Ethical Considerations:** Briefly outline your analysis of the ethical issues in this case, specifically regarding the offer of discounts to minimize food waste.

For Resolution (Recommendations & Limitations):

- Present 2 or 3 recommendations for how CoastalFresh can address food waste. Your recommendations should be based on and supported by your data analysis from Assessments 3 and 4. Avoid general suggestions that are not linked to your analysis.

Hint: You may also make a few relevant external sources to further support your reasoning.

- **Limitations:** summarise any assumptions, limitations, or biases in your data analysis across both Assessments 3 and 4. Explain how these may affect the accuracy of your analysis and the reliability of your recommendations.

Academic Integrity

Using paid academic cheating services to complete your work is strictly prohibited. You may use AI tools for brainstorming ideas initially, but your final submission should reflect your own work.

If there are concerns about who wrote the assignment, the student may be asked to attend a short live interview and explain their work, as per UNSW assessment policy.

UNSW investigates academic misconduct, and **can check it retrospectively**, which means once confirmed, they can change your course grade – even if you completed the course a few years ago or have already graduated. Please do not take the risk.

Get Individual writing feedback on your draft: <https://www.student.unsw.edu.au/feedback-hub>

Get Excel support from our COMM 1110 teachers – [Details](#)

Assessment 4 Marking Criteria

Criteria		Fail	Pass	Credit	Distinction	High Distinction
1. Statistical Data Analysis	40%	Little to no understanding of the dataset. Incorrect or no use of statistical tools, such as no regression output is included Lack of rationale in variable selection. No clear or incorrect correlation or regression analysis.	Basic statistical tools are used, with some small errors. Charts or visuals (such as Linear regression output) are clear, but the explanation may be shallow.	The Excel data analysis is generally accurate, with Excel diagrams that are clear enough to present the results. The analysis includes relevant explanations that address each question, showing a basic understanding of the results. The student reports key findings, with some explanation of patterns or relationships, though this may not be fully developed.	The Excel data analysis is accurate, with clear and well-presented Excel diagrams that are easy to understand and support the results. The analysis includes relevant and well-developed explanations that directly address each question, showing a clear understanding of the results and what they mean. The student goes beyond reporting numbers by explaining patterns, relationships, and key findings in a logical way.	The Excel data analysis is accurate, comprehensive, and supported by clear, relevant diagrams. The student goes beyond simply describing charts or reporting values by offering independent reasoning, critical interpretation, and original insights for each question. The analysis clearly addresses the assessment task and demonstrates a strong understanding of the data's implications from the student's perspective. Data analysis goes beyond simply describing your diagram, table, or data. You need to identify key trends or points from your data and then explain the reasoning behind them from your own perspective. This is the approach managers take when analysing data – it's about interpreting the data in a meaningful way and making decisions based on those insights.

2. Ethics	25%	Fails to apply the correct 7-step framework taught in our course. No clear position on the ethical dilemma.	Basic application of the framework. Clear position on the ethical dilemmas.	Good application of the framework. Clear and well-reasoned position on the ethical dilemma. The framework is applied well, though there is some room for deeper exploration of key points. The position on the ethical dilemma is clear, but the reasoning could be further developed with more specific examples or a broader view of the impact on stakeholders.	Provide a clear and well-structured analysis using the "7-step ethical framework," ensuring each step includes relevant details specific to the case. Focus on key aspects, such as the impact of the ethical dilemma and its potential harm to the key stakeholder. Ensure your analysis is case-specific, rather than overly generic. The analysis should fully address the assessment question, offering thoughtful reasoning, ethical insights, and meaningful reflections from your perspective.	The student provides a clear and well-structured description of a relevant ethical dilemma. The analysis follows the "7-step ethical framework" effectively, with each step clearly addressed. The student goes beyond restating definitions, applying independent ethical reasoning. The use of the framework to analyse the dilemma is thoughtful, offering original insights and demonstrating a deeper understanding of the ethical issues at hand. The analysis fully addresses the assessment question, showcasing the student's ability to navigate the ethical complexity from their perspective, using the "7-step model" to structure their reasoning. Ethical analysis should go beyond simply restating definitions. You need to provide your own insights into the ethical dilemma, considering its broader implications.
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3. Problem Solving	35%	Lack of structure in the overall answer No clear recommendations or solutions. Fails to identify assumptions and limitations.	The overall answer shows a basic structure, but ideas may not always be clearly organised or logically connected. Recommendations or solutions are provided, but they may lack clear support from the analysis or are not fully explained. Some assumptions and limitations are identified, but the explanation is limited or unclear.	The overall answer is generally well-structured, with ideas presented in a mostly logical order. Recommendations are explained and supported by your analysis, showing a basic link between your findings and your proposed solutions, though the reasoning may not be fully developed. Assumptions and limitations are identified, with some explanation of how they may affect your analysis, though the discussion may be brief or not fully detailed.	Clear and well-structured argument throughout the section, with ideas presented in a logical order. Recommendations are clearly explained and supported by your analysis, showing a clear link between your findings and your proposed solutions, with evidence of your own reasoning and judgement. Assumptions and limitations are identified and discussed clearly, with some explanation of how they may affect your analysis and the reliability of your recommendations.	The student fully addresses the assessment question, following all the requirements. They effectively cover the key statistical outcomes from their analysis, presenting the information in a well-structured and clear manner. In their recommendations and limitations, the student demonstrates thoughtful and well-developed reasoning that is clearly relevant to the case. Recommendations are not generic but are directly built on the student's own data analysis, with clear explanation of how their findings are used to support effective and realistic solutions. There are strong evidence of independent thinking and the ability to apply results to the specific context. Limitations are discussed in a deeper and more critical way, clearly explaining how they may affect the results and the strength of the recommendations. The student shows a clear understanding of the reliability of their analysis.
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